

Project Documentation

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Rising Waters – AI Flood Prediction System

1. Introduction

Floods are among the most devastating natural disasters, causing significant damage to life, infrastructure, agriculture, and the economy. Early prediction of flood risk helps authorities and citizens take preventive actions to minimize losses.

The **Rising Waters – AI Flood Prediction System** uses Machine Learning to analyze multiple environmental and geographical factors and predict the probability of flooding. The system provides an easy-to-use web interface built with Flask that generates flood risk predictions along with safety recommendations.

2. Problem Statement

Traditional flood warning systems rely heavily on manual observations and historical records, which can delay decision-making.

This project aims to build an intelligent system capable of predicting flood risk quickly and accurately using machine learning techniques.

3. Objectives

- Develop an AI-based flood prediction model.
- Analyze environmental parameters affecting floods.
- Predict flood probability using Machine Learning.
- Display prediction results through a user-friendly web application.

- Provide flood safety recommendations.
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4. Technologies Used

Programming Language

- Python

Frontend

- HTML5
- CSS3
- JavaScript

Backend

- Flask

Machine Learning

- Scikit-learn
- XGBoost
- Pandas
- NumPy
- Joblib

Database

- SQLite
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5. Dataset

The project uses a Flood Prediction Dataset containing multiple environmental and geographical attributes, including:

- Monsoon Intensity
- Topography Drainage
- River Management

- Deforestation
 - Urbanization
 - Climate Change
 - Dam Quality
 - Siltation
 - Agricultural Practices
 - Encroachments
 - Disaster Preparedness
 - Drainage Systems
 - Watersheds
 - Wetland Loss
 - Planning
 - Political Factors
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6. Machine Learning Workflow

1. Dataset Collection
 2. Data Cleaning
 3. Exploratory Data Analysis
 4. Feature Scaling
 5. Train-Test Split
 6. Model Training
 7. Model Comparison
 8. Best Model Selection
 9. Model Deployment using Flask
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7. Models Implemented

- Decision Tree
- Random Forest
- K-Nearest Neighbors (KNN)
- XGBoost

The XGBoost model provided the best prediction performance and was selected for deployment.

8. Application Features

- Predict flood probability
- Display flood risk level
- Safety recommendations
- Responsive web interface
- Animated rainfall background
- Reset functionality
- SQLite database integration

9. System Architecture

User

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Flask Web Application

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Input Validation

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Feature Scaling

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XGBoost Model

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Prediction Result

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Safety Recommendations

10. Results

The application successfully predicts flood risk based on user inputs and displays:

- Flood probability
- Risk level (Low / Moderate / High)
- Safety recommendations

The system provides predictions within a few seconds through an interactive interface.

11. Future Enhancements

- Live Weather API Integration
- Real-Time Rainfall Monitoring
- SMS and Email Alerts
- Satellite Image Analysis
- Interactive GIS Maps
- Cloud Deployment
- Mobile Application

12. Conclusion

The Rising Waters AI Flood Prediction System demonstrates how Machine Learning can assist in early flood risk prediction. By combining environmental parameters with intelligent algorithms, the application provides quick and reliable predictions that can support disaster preparedness and improve public safety.